SANKET GOMEKAR

3800 SW, 34th Street, Apt BB273, Gainesville, FL-32608 +1(352)-888-1096
sanket.gomekar@gmail.com http://www.linkedin.com/in/sanketgomekar

Areas of interest: Product Design, Automobile Engineering, Manufacturing, Product Development, Testing & Validation, Finite Element Analysis

EDUCATION:

- **Masters of Science**, Mechanical Engineering with specialization in Solid Mechanics, Design & Manufacturing **May 2018**
  University of Florida, Gainesville, Florida **GPA:** 3.55/4.0
- **Bachelor of Engineering (B.E.),** Mechanical Engineering **July 2016**
  Nagpur University, Nagpur, India **GPA:** 3.8/4.0

RELEVANT COURSEWORK: Applied Elasticity, Structural optimization, Non-linear FEA, Automotive engineering, Supply Chain Management

TECHNICAL SKILLS: CATIA (Intermediate), ANSYS (Intermediate), Abaqus (Intermediate), Microsoft Office (Word, Excel, PowerPoint)

RELEVANT EXPERIENCE:

- **Tesla Inc, Palo Alto, California** **August 2017 – December 2017**
  Fall Intern: Materials and Fastener Engineering
  1. Designed a test fixture and setup for finding out the clamp load in riveted joints
  2. Tested various new and existing fasteners for benchmarking, performance and/or validating supplier data
  3. Co-ordinated with multiple teams to recommend fasteners based on application.
  4. Collaborated with suppliers to support fastener requirements.
- **SAE BAJA, Team Impulsifiers** **April 2014 - February 2016**
  Lead Engineer: Hydraulic Braking System Design and Manufacturing
  1. Led the hydraulic braking system design and manufacturing team of 4 members for 2 National BAJA competitions.
  2. Improved the braking performance than previous year's vehicle.
  3. Assisted in overall chassis manufacturing and gained hands on experience in cutting, drilling, milling, grinding, & welding.
- **NTPC Ltd, Korba, Chattisgarh, India** **June 2015 – July 2015**
  Summer Intern Trainee: power Plant Engineering
  1. Scrutinized the overhauling of 500 MW unit
  2. Analyzed the structure and working of two most important components of thermal power plant: boilers and turbines

PROJECTS:

- **Simulation of deep drawing process and spring back effect** **April 2017**
  1. Modelled and simulated the deep drawing process of a metal blank
  2. Two different elements were used for modelling
  3. Results were compared for the different cases (hollow and solid elements, different clamping forces)
- **Structural Optimization:** *Topology and sizing optimization for optimum shape of structure* **March 2017**
  1. Used topology optimization for initial conceptual design of the structure (using CALFEM)
  2. Used sizing optimization for determining the final, feasible shape of the structure (fully stressed design)
- **Non-linear finite element analysis for hyperelastic material** **March 2017**
  1. Determined hyperelastic material properties by curve fitting the experimental stress values of the material
  2. Non-linear FEA of rubber gasket assuming hyperelastic material
  3. Suggested modifications in geometry of rubber gasket by observing results of FEA
- **Senior Year Design Project:** *Open air mist cooling system* **May 2015 – April 2016**
  1. Designed and manufactured a solar power operated mist cooling system.
  2. Analysed performance of the system by varying parameters (number of nozzles, nozzle height, air draft and wind speed)
  3. Determined the values of parameters for optimum performance and desired cooling effect.
  4. Performed calculations to run the mist cooling system on solar power